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Essentials of Data Science
Laboratory - 2304102L - 2026

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1. Practical 1

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5.1. Practice Lab Assignment

5.1.1 Stacked Plot

5.2 Lab Assignment

5.1.1. Stacked Plot

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

Create a stacked area plot using the data.

Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.

Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

1

import matplotlib.pyplot as plt

2

import pandas as pd

3

4

Data for Months and Temperature for three cities

5

data = {

6

'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September', 'October', 'November', 'December'],

7

'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],

8

'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],

9

'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]

10

}

11

Write your code...

12

import matplotlib.pyplot as plt

13

import pandas as pd

14

15

16

Data for Months and Temperature for three cities

17

data = {

18

'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September', 'October', 'November', 'December'],

19

'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],

20

'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],

21

'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]

22

}

23

24

25

26

27

28

29

30

31

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5.2.1. Titanic Dataset

00:42

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the pandas library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.

titanicDat...

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1# import pandas as pd

2# import matplotlib.pyplot as plt

3

4## Load the Titanic dataset from the CSV file

5# df = pd.read_csv('titanic.csv')

6

7## Set up the figure for 5 subplots

8# fig, axes = plt.subplots(3, 2, figsize=(12, 12))

9

10## write the code..

11import pandas as pd

12import matplotlib.pyplot as plt

13

14# Load the Titanic dataset from the CSV file

15df = pd.read_csv('titanic.csv')

16

17# Set up the figure for 5 subplots

18fig, axes = plt.subplots(3, 2, figsize=(12, 12))

19

20## write the code..

21count=df['Pclass'].value_counts().sort_index()

22count_gen=df['Gender'].value_counts()

23count_s=df['Survived'].value_counts().sort_index()

24

25axes[0,0].bar(count.index,count.values,color='skyblue')

TerminalTest cases

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5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**.
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below.

[illegible]

Sample Data:

PassengerId,Survived,Lastname,Firstname,Sex,Age, SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.92,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373456,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,1,"Maloney, Martin Joseph",male,27,1,2,2,240002,21.075,C

Histogra... Submit

```
1 # import pandas as pd
2 # import matplotlib.pyplot as plt
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6
7 ## Data Cleaning
8 # data['Age'].fillna(data['Age'].median(), inplace=True)
9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 # data.drop('Cabin', axis=1, inplace=True)
11
12 ## Convert categorical features to numeric
13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # import pandas as pd
17 # import matplotlib.pyplot as plt
18
19 ## Load the dataset (replace 'titanic.csv' with your file path)
20
21
22 ## Drop missing values in the 'Age' column
23 # age_data = df['Age'].dropna()
24
25 # # Plot histogram
```

5.2.3. Bar plot of survival rate of passengers

08:12

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).

2. Set the chart type to 'bar'.

3. Add the title "Survival Count" to the chart.

4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Berthelme, Master. Gustave Louis",male,2,3,3,740000,51.075,C

BarPlotOf...

Submit

1 # import pandas as pd

2 # import matplotlib.pyplot as plt

3

4 ## Load the Titanic dataset

5 # data = pd.read_csv('Titanic-Dataset.csv')

6

7 ## Data Cleaning

8 # data['Age'].fillna(data['Age'].median(), inplace=True)

9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 # data.drop('Cabin', axis=1, inplace=True)

11

12 ## Convert categorical features to numeric

13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 ## Write your code here for Bar Plot for Survival Rate

17

18 # data['Survived'].value_counts().sort_index().plot(kind='bar')

19

20 # plt.title('Survival Count')

21 # plt.xlabel('Survived')

22 # plt.ylabel('Count')

23

24 # plt.show()

25 import pandas as pd

Terminal

Test cases

5.2.4. Bar Plot for Survival by Gender

08:05

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

- Group the data by the 'Sex' column, then use the `value_counts()` function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
- Use a **stacked bar chart** to display the survival counts.
- Add the title **"Survival by Gender"** to the chart.
- Label the x-axis as 'Gender' and the y-axis as 'Count'.
- The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
```

BarPlotOf...

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```
1 # import pandas as pd
2 # import matplotlib.pyplot as plt
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6
7 ## Data Cleaning
8 # data['Age'].fillna(data['Age'].median(), inplace=True)
9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 # data.drop('Cabin', axis=1, inplace=True)
11
12 ## Convert categorical features to numeric
13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 ## Write your code here for Bar Plot for Survival by Gender
17 # plt.figure()
18 # data.groupby('Sex')
19 # ['Survived'].value_counts().unstack().plot(kind='bar', stacked=True)
20 # plt.title('Survival by Gender')
21 # plt.xlabel('Gender')
22 # plt.ylabel('Count')
23 # plt.legend(['Not Survived', 'Survived'])
24 # plt.show()
25 import pandas as pd
```

Terminal Test cases

5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts().

2. Use a stacked bar chart to display the survival counts.

3. Add the title "Survival by Pclass" to the chart.

4. Label the x-axis as 'Pclass' and the y-axis as 'Count'.

5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

BarPlotOf...

1 # import pandas as pd

2 # import matplotlib.pyplot as plt

3

4 ## Load the Titanic dataset

5 # data = pd.read_csv('Titanic-Dataset.csv')

6

7 ## Data Cleaning

8 # data['Age'].fillna(data['Age'].median(), inplace=True)

9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 # data.drop('Cabin', axis=1, inplace=True)

11

12 ## Convert categorical features to numeric

13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 ## Write your code here for Bar Plot for Survival by Pclass

17

18

19 import pandas as pd

20 import matplotlib.pyplot as plt

21

22 # Load the Titanic dataset

23 data = pd.read_csv('Titanic-Dataset.csv')

24

25 # Data Cleaning

Terminal

Test cases

5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.
The chart should display the following specifications:
1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "**Survival by Embarked**" to the chart.
4. Label the x-axis as '**Embarked**' and the y-axis as '**Count**'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

BarPlotOf...

1 # import pandas as pd
2 # import matplotlib.pyplot as plt
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6
7 ## Data Cleaning
8 # data['Age'].fillna(data['Age'].median(), inplace=True)
9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 # data.drop('Cabin', axis=1, inplace=True)
11
12 ## Convert categorical features to numeric
13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 ## Write your code here for Bar Plot for Survival by Embarked
17 # import pandas as pd
18 # import matplotlib.pyplot as plt
19
20 ## Load dataset (replace 'titanic.csv' with your file path)
21 # df = pd.read_csv('titanic.csv')
22
23 ## Convert Embarked column into dummy variables
24 # embarked_dummies = pd.get_dummies(df['Embarked'], prefix='Embarked')
25

Terminal Test cases

5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below.

[illegible]

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Halkinnen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.92,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373456,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,330677,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.0625,E46,S
8,0,3,"O'Brien, Miss. Constance",female,23,3,3,340000,31.075,C

```

1  ## import pandas as pd
2  ## import matplotlib.pyplot as plt
3
4  ### Load the Titanic dataset
5  ## data = pd.read_csv('Titanic-Dataset.csv')
6
7  ### Data Cleaning
8  ## data['Age'].fillna(data['Age'].median(), inplace=True)
9  ## data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 ## data.drop('Cabin', axis=1, inplace=True)
11
12 ### Convert categorical features to numeric
13 ## data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 ## data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 ### Write your code here for Box Plot for Age by Pclass
17
18
19 import pandas as pd
20 import matplotlib.pyplot as plt
21
22 ## Load the Titanic dataset
23 ## data = pd.read_csv('Titanic-Dataset.csv')
24
25 ## Data Cleaning

```


5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).

2. Set the title of the plot to **"Age by Survival"**.

3. Remove the default subtitle with **plt.suptitle("")**.

4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S	
6	0	3	Noran, Mr. James	male	0	0	0	330877	8.4583	Q	

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Noran, Mr. James",male,0,0,330877,8.4583,,Q

BoxPlotF...

1

import pandas as pd

2

import matplotlib.pyplot as plt

3

4

Load the Titanic dataset

5

data = pd.read_csv('Titanic-Dataset.csv')

6

7

Data Cleaning

8

data['Age'].fillna(data['Age'].median(), inplace=True)

9

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10

data.drop('Cabin', axis=1, inplace=True)

11

12

Convert categorical features to numeric

13

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16

Write your code here for Box Plot for Age by Survived

17

18

19

import pandas as pd

20

import matplotlib.pyplot as plt

21

22

Load the Titanic dataset

23

data = pd.read_csv('Titanic-Dataset.csv')

24

25

Data Cleaning

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Test cases

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5.2.9. Box Plot for Fare by Pclass 00:37 🔍 🌙 📝 🔗 ⌵

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **"Pclass"** and the y-axis as **"Fare"**.

The Titanic dataset contains columns as shown below.

[illegible]

Sample Data:

```

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171/7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599/71.2833,C85,C
3,1,3,"Hulkkins, Miss. Laina",female,26,0,0,STON/O2. 3101283/7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803/53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450/8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877/8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463/51.8625,E46,S
8,0,2,"Palsson, Martin Gustaf",male,21,2,3,349006/24.075,C

```

BoxPlotF... Submit

```

1 # import pandas as pd
2 # import matplotlib.pyplot as plt
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6
7 ## Data Cleaning
8 # data['Age'].fillna(data['Age'].median(), inplace=True)
9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 # data.drop('Cabin', axis=1, inplace=True)
11
12 ## Convert categorical features to numeric
13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 ## Write your code here for Box Plot for Fare by Pclass
17
18
19 import pandas as pd
20 import matplotlib.pyplot as plt
21
22 # Load the Titanic dataset
23 data = pd.read_csv('Titanic-Dataset.csv')
24
25 # Data Cleaning

```

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5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

- Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
- Set the title of the plot to **"Age vs. Fare"**.
- Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

AgeFareS...

Submit

```
1 # import pandas as pd
2 # import matplotlib.pyplot as plt
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6
7 ## Data Cleaning
8 # data['Age'].fillna(data['Age'].median(), inplace=True)
9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 # data.drop('Cabin', axis=1, inplace=True)
11
12 ## Convert categorical features to numeric
13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 ## Write your code here for Box Plot for Fare by Pclass
17
18
19 import pandas as pd
20 import matplotlib.pyplot as plt
21
22 # Load the Titanic dataset
23 data = pd.read_csv('Titanic-Dataset.csv')
24
25 # Data Cleaning
```

Terminal Test cases

5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.

2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).

3. Set the title of the plot to **"Age vs. Fare by Survival"**.

4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

AgeFareS...

Submit

```
1 # import pandas as pd
2 # import matplotlib.pyplot as plt
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6
7 ## Data Cleaning
8 # data['Age'].fillna(data['Age'].median(), inplace=True)
9 # data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 # data.drop('Cabin', axis=1, inplace=True)
11
12 ## Convert categorical features to numeric
13 # data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 ## Write your code here for Scatter Plot for Age vs. Fare by Survived
17
18
19 import pandas as pd
20 import matplotlib.pyplot as plt
21
22 # Load the Titanic dataset
23 data = pd.read_csv('Titanic-Dataset.csv')
24
25 # Data Cleaning
```

Terminal Test cases